



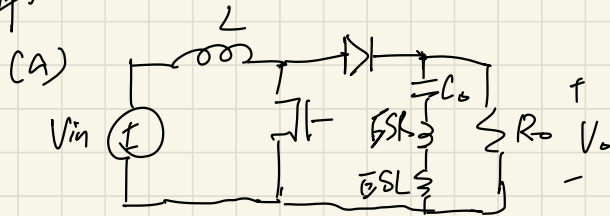
1. (45%) Lecture PEIC-2 p.47~49 group discussion related questions (total 3 questions)

(a) V_o Waveform of boost converter if C_o has ESR and ESL?

(b) Give three reasons why **boost** converter $V_o \geq V_{in}$.

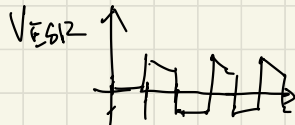
(c) Derivation of DCM Boost Converter voltage gain as a function of duty cycle and load resistance

解:



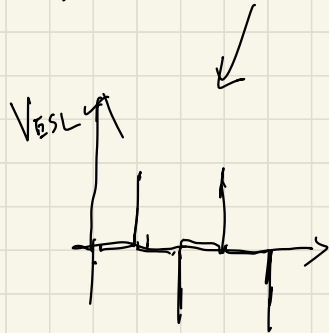
ESR: $V = IR$

ESL: $V = L \frac{di}{dt}$

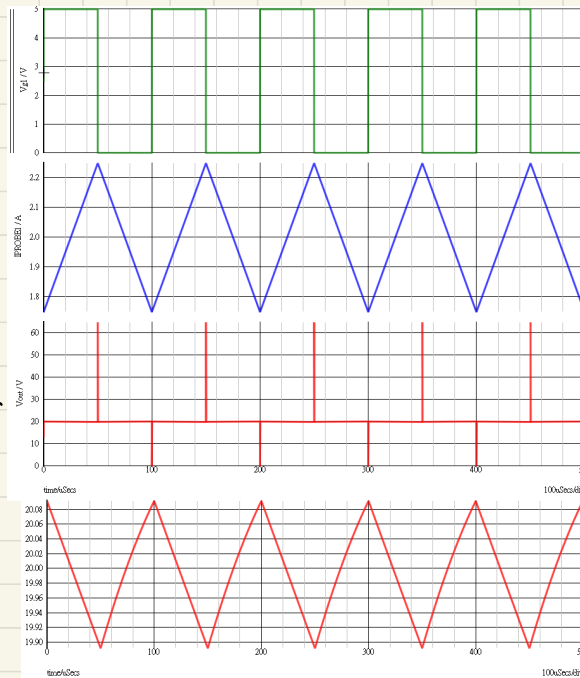


non-ideal cap

ideal cap



there will be spike at edge



(b) ① $\frac{V_o}{V_{in}} = \frac{1}{1-D} \because 0 < D < 1 \rightarrow V_o \geq V_{in}$

② $P_{in} = P_{out}$

$\therefore \langle I_{in} \rangle > \langle I_o \rangle$

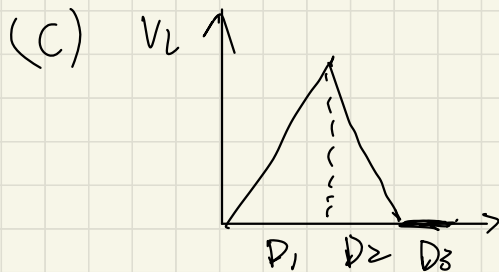
$\therefore \langle V_{in} \rangle < \langle V_o \rangle$

③ Volt-second Balance

$V_{in} \cdot D + (V_{in} - V_o)(1-D) = 0$

$V_{in} \cdot D > 0 \rightarrow (V_{in} - V_o)(1-D) < 0$

$V_{in} < V_o \quad \#$

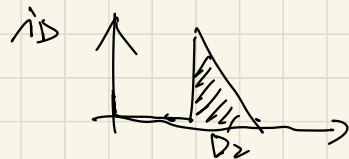


$D_1 V_{in} + D_2 (V_{in} - V_{out}) + D_3 \cdot 0 = 0$

$\langle \dot{i}_D \rangle = \langle \dot{i}_C \rangle + \langle \dot{i}_O \rangle$

$\langle \dot{i}_D \rangle = \langle \dot{i}_O \rangle$

$D_2 = \frac{V_{in}}{V_{out} - V_{in}} D_1$



$\frac{\frac{V_{in}}{L} D_1 T \times D_2 T}{2T} = \frac{V_{out}}{R}$

$\frac{D_1 D_2 V_{in}}{2L} = \frac{V_{out}}{R}$

$$\frac{V_o}{V_{in}} = \frac{RD_2 T}{2L} \quad \frac{RD_1^2 \frac{V_{in}}{V_o V_{in}} T}{2L}$$

$$= \frac{1 \pm \sqrt{1 + \frac{2RD_1^2 T}{L}}}{2}$$

$$= \frac{1 \pm \sqrt{1 + \frac{4D_1^2}{K}}}{2} \quad K = \frac{RT}{2L} \#$$

$$\frac{V_o}{V_{in}} = \frac{RD_1^2 \frac{V_o}{V_{in}} T}{2L}$$

$$2LX = RD_1^2 \frac{1}{X-1}$$

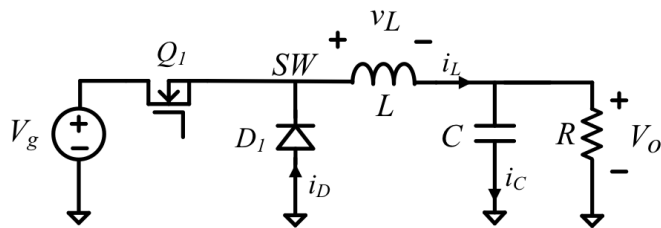
$$2LX^2 - 2LX = RD_1^2 T$$

$$2LX^2 - 2LX - RD_1^2 T$$

$$X = \frac{2L \pm \sqrt{4L^2 + 8L RD_1^2 T}}{4L}$$

$$\frac{V_o}{V_{in}} = \frac{1 \pm \sqrt{1 + \frac{2RD_1^2 T}{L}}}{2}$$

2. A Buck converter in steady-state operation



- (a) (25%) Critical resistance is defined as the value of load resistance R , which achieves operation at the CCM/DCM boundary. Assume $V_g = 10$ V, $V_o = 0.5$ V, Diode D_1 has a forward voltage of 0.5 V, $L = 10$ μ H, $C = 100$ μ F, and Switching frequency $f_s = 2$ MHz, where R is 10% of critical resistance.

Derive critical resistance value, peak and valley values of V_o , i_L , i_C , SW , and duty cycle D .

- (b) (30%) Simulate the converter in (a) to get the steady-state time waveform as p. 42 in week 2 lecture slide. Paste the simulated circuit picture and the simulated waveforms. Mark the important values such as peak and valley currents/voltages and D to prove your calculation in (a). Discuss its difference with your calculation.

解:

(a)

$$V_g = 10 \text{ V}$$

$$V_o = 0.5 \text{ V}$$

$$\frac{V_g - V_o}{L} DT + \frac{0.5 - V_o}{L} (1-D)T = 0$$

$$(10 - 0.5) D = (1-D)$$

$$9.5 D = 1 - D$$

$$10.5 D = 1$$

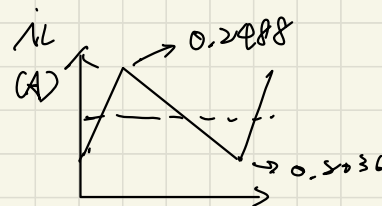
$$D = 0.09524$$

$$R_{crit} = \frac{V_{in} - V_o}{L} DT = 2 \langle i_L \rangle = 2 \langle i_o \rangle = \frac{2V_o}{R}$$

$$R_{crit} = \frac{2V_o}{(V_{in} - V_o) DT} = 22.105 \Omega$$

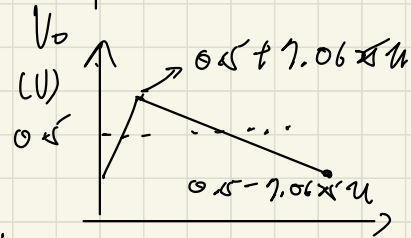
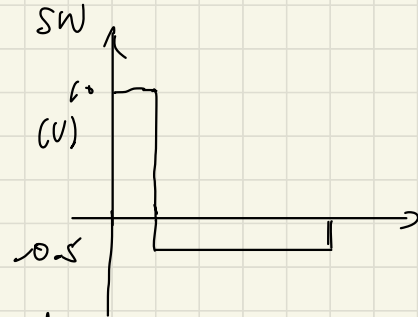
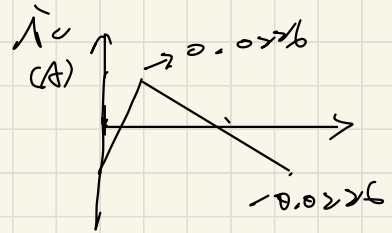
$$R = 2.2105 \Omega$$

$$\langle i_L \rangle = \frac{V_o}{R} = 0.2262 \text{ A}$$



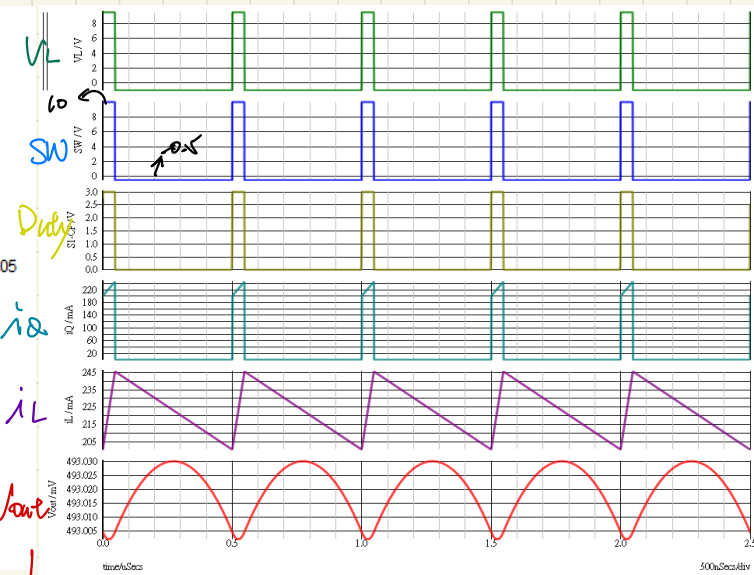
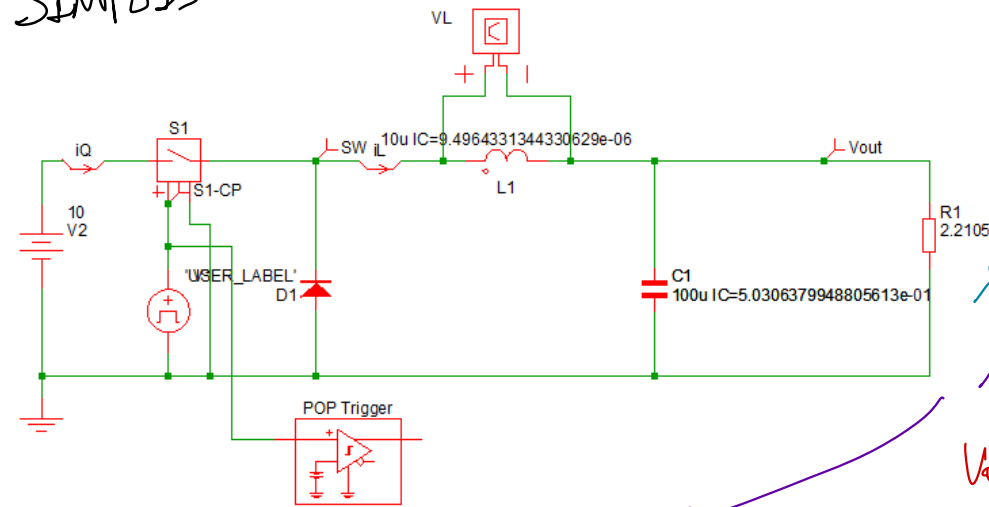
$$\Delta \hat{I}_L = \frac{V_{in} - V_0}{L} \Delta T \times \frac{1}{2} = 0.0226$$

$$\Delta \hat{I}_C = \Delta \hat{I}_L$$



$$\Delta V_0 = \frac{\Delta I_L}{8f_s C} = 1.4125 \times 10^{-5}$$

SIMPLIS



avg = 523mA
 ripple = 22mA
 Similar to
 calculation

avg $\downarrow$ 493 mV
 ripple = ± 0.015 m (differ due to power loss in component)